Handout Matrix Algebra

Basic Matrix Algebra relations

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1 Scalars, Vectors and Matrices

1.1 Notation

let

$$\boldsymbol{A} = \begin{bmatrix} a_{11} & \cdots & a_{1m} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nm} \end{bmatrix}, \text{ and } \boldsymbol{a}_j = \begin{pmatrix} a_{1j} \\ \vdots \\ a_{nj} \end{pmatrix}, \; j = 1, \ldots, m,$$

then the matrix $\boldsymbol{A} = (\boldsymbol{a}_1, \ldots, \boldsymbol{a}_m)$ is of order (also referred to as size or dimension) $n \times m$, and the vectors $\boldsymbol{a}_j$, $j = 1, \ldots, m$, are n-vectors, i.e. $n \times 1$. We also write $\boldsymbol{A} = \{a_{ij}\}$ and $A_{ij} = a_{ij}$.

1.2 Simple definitions

$\boldsymbol{A}$ is *square* if $n = m$.
$\boldsymbol{A}$ is *symmetric* if $n = m$ and $a_{ij} = a_{ji}$, $i, j = 1 \ldots n$.
$\boldsymbol{A}$ is *diagonal* if $n = m$ and $a_{ij} = 0$ if $i \neq j$, $i, j = 1 \ldots n$.
$\boldsymbol{I}_n$ is the $n \times n$ *identity matrix* if it is diagonal with ones on the diagonal.

The column vectors of $\boldsymbol{I}_n = (\boldsymbol{e}_1, \ldots, \boldsymbol{e}_n)$ are *unit* vectors. So, $\boldsymbol{e}_j$ consists of zeros and a single one on the jth position. The size should follow from the context.

Another frequently used vector consists of ones only $\boldsymbol{\imath}_n$. The index indicates the size. Without index, $\boldsymbol{\imath}$, the size should follow from the context. A vector of zeros is denoted by $\mathbf{0}$ and a matrix of zeros by $\mathbf{O}$.

1.3 Simple operations

The *transpose* of a matrix $\boldsymbol{A} = \{a_{ij}\}$ is given by $\boldsymbol{A}' = \{a_{ji}\}$. It is of order $m \times n$.

Result 1 $\boldsymbol{A}$ *is symmetric* $\Leftrightarrow \boldsymbol{A} = \boldsymbol{A}'$.

Remark 1 *By the way, instead of the symbol* $\Leftrightarrow$ *also the phrase* if and only if *could have been used, which is sometimes abbreviated to* iff. *That is to say, if condition* $\mathcal{A}$ implies *condition* $\mathcal{B}$*, then we say* if $\mathcal{A}$ holds then $\mathcal{B}$ holds*, or we say,* $\mathcal{A}$ is sufficient for $\mathcal{B}$*, or we say* $\mathcal{B}$ is necessary for $\mathcal{A}$*. This is denoted by* $\mathcal{A} \Rightarrow \mathcal{B}$*. If* $\mathcal{A}$ *is necessary and sufficient for* $\mathcal{B}$*, then we write* $\mathcal{A} \Leftrightarrow \mathcal{B}$*. In that case* $\mathcal{A}$ *and* $\mathcal{B}$ *are equivalent. Notice that*

the statement $\mathcal{A} \Rightarrow \mathcal{B}$ *is equivalent to* $\sim \mathcal{B} \Rightarrow \sim \mathcal{A}$*, where* $\sim$ *indicates* not*. Furthermore, if* $A = \{a | a \text{ satifies condition } \mathcal{A}\}$ *indicates the set of elements* a *that satisfy condition* $\mathcal{A}$*, and* $B = \{a | a \text{ satifies condition } \mathcal{B}\}$*, then* $\mathcal{A} \Rightarrow \mathcal{B}$ *is equivalent to* $A \subset B$*, or, if* $\bar{A}$ *is the complement of* A*,* $\bar{B} \subset \bar{A}$*.*

The *summation* of two $n \times m$ matrices $\boldsymbol{A}$ and $\boldsymbol{B}$ is given by $\boldsymbol{A} + \boldsymbol{B} = \{a_{ij} + b_{ij}\}$.

The *multiplication* $\boldsymbol{AB}$ of an $n \times m$ matrix $\boldsymbol{A}$ and an $m \times k$ matrix $\boldsymbol{B}$ is given by $\{\boldsymbol{AB}\}_{ij} = \{\sum_{l=1}^{m} a_{il} b_{lj}\}$. Notice the sizes of the matrices should be such that the number of columns of the first matrix equals the number of rows of the second matrix.

Remark 2 *The number of rows of* $\boldsymbol{ABCDEFG}$ *equals the number of rows of* $\boldsymbol{A}$*, and its number of columns equals the number of columns of* $\boldsymbol{G}$*. If* $\boldsymbol{G}$ *is a standing vector, i.e.* $\boldsymbol{G}$ *has one column, then so is* $\boldsymbol{ABCDEFG}$*. If* $\boldsymbol{A}$ *is a lying vector, i.e.* $\boldsymbol{A}$ *has one row, then so is* $\boldsymbol{ABCDEFG}$*. So, the question whether* $\boldsymbol{ABCDEFG}$ *is either a scalar, a vector or a matrix depends only on the number of rows of* $\boldsymbol{A}$ *and the number of columns of* $\boldsymbol{G}$*.*

Remark 3 *Notice that premultiplication of* $\boldsymbol{A}$ *by a diagonal matrix* $\boldsymbol{D}$*, has the effect that the rows of* $\boldsymbol{A}$ *are mulptiplied by the diagonal elements of* $\boldsymbol{D}$*. Postmultiplication affects the columns.*

Sometimes sizes are assumed implicitly to be conformable, for example $\boldsymbol{e}_i' \boldsymbol{A} \boldsymbol{e}_j = a_{ij}$, where it is assumed $\boldsymbol{e}_i$ is the ith column of $\boldsymbol{I}_n$ and $\boldsymbol{e}_j$ is the jth column of $\boldsymbol{I}_m$.

Result 2 *For conformable matrices,* $(\boldsymbol{ABC})' = \boldsymbol{C}'\boldsymbol{B}'\boldsymbol{A}'$*.*

Some very important multiplications always fit.

Result 3 $$\boldsymbol{AA}' = \sum_{j=1}^{m} \boldsymbol{a}_j \boldsymbol{a}_j'.$$

Result 4 *If* $\boldsymbol{A}$*,* $\boldsymbol{B}$ *and* $\boldsymbol{D}$ *have* m *columns and* $\boldsymbol{D}$ *is diagonal then* $\boldsymbol{ADB}' = \sum_{j=1}^{m} d_j \boldsymbol{a}_j \boldsymbol{b}_j'$*, where* $\boldsymbol{a}_j$ *and* $\boldsymbol{b}_j$ *are (standing) vectors and the* d_j *are scalars,* $j = 1, \ldots, m$*.*

2 Rank, nonsingularity and inverses

2.1 Linear dependencies

The $n \times m$ matrix $\boldsymbol{A}$ has *full column rank* m if for an m-vector $\boldsymbol{x}$ holds that $\boldsymbol{Ax} = \boldsymbol{0} \Rightarrow \boldsymbol{x} = \boldsymbol{0}$. In that case the m columns of $\boldsymbol{A}$ are *linearly independent.* The *rank* of a matrix is equal to the maximum number of linearly independent columns of the matrix.

Result 5 $$\text{rank}(\boldsymbol{A}) = \text{rank}(\boldsymbol{A}').$$

Result 6 *If the $n \times m$ matrix A has full column rank, then $n \geq m$.*

Result 7 $$\text{rank}(\boldsymbol{BA}) \leq \text{rank}(\boldsymbol{A}).$$
(Since $\boldsymbol{Ax} = \boldsymbol{0} \Rightarrow \boldsymbol{BAx} = \boldsymbol{0}$.)

Result 8 *If $\boldsymbol{C}$ has full column rank, then* $\text{rank}(\boldsymbol{CA}) = \text{rank}(\boldsymbol{A})$.
(Since $\boldsymbol{CAx} = \boldsymbol{0} \Rightarrow \boldsymbol{Ax} = \boldsymbol{0}$.)

Suppose $y = Ax \neq 0$, then $Cy = 0$, while $y \neq 0$. This is impossible since C has f.c.r.

Result 9 $$\boldsymbol{Ax} = \boldsymbol{0} \Leftrightarrow \boldsymbol{x}'\boldsymbol{A}'\boldsymbol{Ax} = 0.$$

Exercise 1 *Prove Result 9.*

Result 10 $$\boldsymbol{x}'\boldsymbol{A}'\boldsymbol{Ax} = 0 \Leftrightarrow \boldsymbol{A}'\boldsymbol{Ax} = \boldsymbol{0}.$$

Exercise 2 *Prove Result 10.*

Result 11 $$\text{rank}(\boldsymbol{A}'\boldsymbol{A}) = \text{rank}(\boldsymbol{A}).$$
(Since $\boldsymbol{A}'\boldsymbol{Ax} = \boldsymbol{0} \Leftrightarrow \boldsymbol{Ax} = \boldsymbol{0}$.)

Result 12 $$\text{rank}(\boldsymbol{AA}') = \text{rank}(\boldsymbol{A}).$$

Exercise 3 *Prove Result 12.*

2.2 Nonsingularity

$\boldsymbol{C}$ is said to be *nonsingular* if it is square with full rank. If $\boldsymbol{C}$ is nonsingular, then there exists a matrix $\boldsymbol{C}^{-1}$ such that $\boldsymbol{CC}^{-1} = \boldsymbol{C}^{-1}\boldsymbol{C} = \boldsymbol{I}$. The matrix $\boldsymbol{C}^{-1}$ is called the *inverse* of $\boldsymbol{C}$ and it is unique.

Result 13 *$\boldsymbol{C}^{-1}$ is nonsingular.*

Exercise 4 *Prove Result 13.*

Result 14 *If $\boldsymbol{C}$ is square and $\boldsymbol{BC} = \boldsymbol{I}$, then $\boldsymbol{B} = \boldsymbol{C}^{-1}$.*

Proof: As $\boldsymbol{BC} = \boldsymbol{I}$, we only have to prove $\boldsymbol{CB} = \boldsymbol{I}$. As $\boldsymbol{C}$ is square and $\boldsymbol{BC} = \boldsymbol{I}$, we know from Result 7 that both $\boldsymbol{B}$ and $\boldsymbol{C}$ must be nonsingular. Let $\boldsymbol{CB} = \boldsymbol{A}$ and premultiply both sides by $\boldsymbol{B}$. We find, using $\boldsymbol{BC} = \boldsymbol{I}$, that $\boldsymbol{B} = \boldsymbol{BA}$, so that $\boldsymbol{B}(\boldsymbol{I} - \boldsymbol{A}) = \boldsymbol{O}$. As $\boldsymbol{B}$ is nonsingular we find $\boldsymbol{A} = \boldsymbol{I}$. q.e.d.

Result 15 *If $\boldsymbol{A}$, $\boldsymbol{B}$, and $\boldsymbol{C}$ are nonsingular matrices, then* $(\boldsymbol{ABC})^{-1} = \boldsymbol{C}^{-1}\boldsymbol{B}^{-1}\boldsymbol{A}^{-1}$.

Result 16 *If $\boldsymbol{A}$ is nonsingular, then* $(\boldsymbol{A}')^{-1} = (\boldsymbol{A}^{-1})'$.

Exercise 5 *Prove Result 16.*

Exercise 6 *Let $\boldsymbol{A}$ be nonsingular and $\boldsymbol{b}'\boldsymbol{A}\boldsymbol{b} \neq -1$. Prove*

$$(\boldsymbol{A} + \boldsymbol{b}\boldsymbol{b}')^{-1} = \boldsymbol{A}^{-1} - \frac{\boldsymbol{A}^{-1}\boldsymbol{b}\boldsymbol{b}'\boldsymbol{A}^{-1}}{1 + \boldsymbol{b}\boldsymbol{A}^{-1}\boldsymbol{b}}.$$

2.3 Normalized vectors

The (Euclidean) *length* (or *norm*) of a vector $\boldsymbol{a}$ is given by $\|\boldsymbol{a}\| = \sqrt{\boldsymbol{a}'\boldsymbol{a}}$. The column vectors of a matrix are *orthogonal* if $\boldsymbol{A}'\boldsymbol{A}$ is diagonal. The matrix $\boldsymbol{A}$ is *orthonormal* if $\boldsymbol{A}'\boldsymbol{A} = \boldsymbol{I}$, so that the columns are both orthogonal and normalized.

Result 17 *If $\boldsymbol{P}$ is orthonormal and square, then* $\boldsymbol{P}' = \boldsymbol{P}^{-1}$.

As $P'P = I$ and P is square the result follows from Res.

The proof follows from Result 14. So, we find that if $\boldsymbol{P}$ is square and $\boldsymbol{P}'\boldsymbol{P} = I$, then $\boldsymbol{P}\boldsymbol{P}' = I$, i.e. the rows are orthogonal and normalized as well.

3 Decompositions and eigenvalues

week 1.1

3.1 Singular value decomposition

Let $\boldsymbol{A}$ be an $n \times m$ matrix with $\text{rank}(\boldsymbol{A}) = r$. The *Singular Value Decomposition (SVD)* of $\boldsymbol{A}$ is given by:

$$\boldsymbol{A} = \boldsymbol{PDQ}' = \sum_{i=1}^{r} d_i \boldsymbol{p}_i \boldsymbol{q}_i',$$

(as in Result 4), where $\boldsymbol{P}$ is an $n \times r$ orthonormal matrix, $\boldsymbol{P} = (\boldsymbol{p}_1, \ldots, \boldsymbol{p}_r)$, $\boldsymbol{Q}$ is an $m \times r$ orthonormal matrix, $\boldsymbol{Q} = (\boldsymbol{q}_1, \ldots, \boldsymbol{q}_r)$, and $\boldsymbol{D}$ is a diagonal $r \times r$ matrix with diagonal elements $d_1 \geq d_2 \geq \ldots \geq d_r > 0$, called *singular values*. So $\boldsymbol{A}$ can be decomposed as a summation of r matrices of rank one whose columns and rows are orthogonal.

3.2 Eigenvalue-eigenvector decomposition of a symmetric matrix

Let $\boldsymbol{S}$ be symmetric $n \times n$, then the *spectral decomposition*, or *eigenvalue decomposition* is given by $\boldsymbol{S} = \boldsymbol{PDP}'$, where $\boldsymbol{P}$ is a square orthonormal matrix, and $\boldsymbol{D}$ is diagonal. The column vectors of $\boldsymbol{P} = (\boldsymbol{p}_1, \ldots, \boldsymbol{p}_n)$ are *eigenvectors* and the diagonal elements of $\boldsymbol{D}$ are called *eigenvalues*, or *latent roots* or *characteristic roots*. That is to say $\boldsymbol{Sp}_i = \boldsymbol{p}_i d_i$, for $i = 1, \ldots, n$. We also have $\boldsymbol{P}'\boldsymbol{P} = \boldsymbol{PP}' = I_n$. The eigenvalues need not all be positive.

The SVD of a symmetric matrix $\boldsymbol{S}$ finds essentially the same solution. If the eigenvalue d_i is negative, then the singular value is given by $|d_i|$ and $\boldsymbol{q}_i = -\boldsymbol{p}_i$. Furthermore, $n - r$ eigenvalues are equal to zero.

Result 18 *Let $\boldsymbol{S}$ be symmetric. The largest and smallest eigenvalues of $\boldsymbol{S}$ are given by*

$$\lambda_{max} = \max_{\boldsymbol{x}} \frac{\boldsymbol{x}'\boldsymbol{Sx}}{\boldsymbol{x}'\boldsymbol{x}}, \qquad \lambda_{min} = \min_{\boldsymbol{x}} \frac{\boldsymbol{x}'\boldsymbol{Sx}}{\boldsymbol{x}'\boldsymbol{x}},$$

respectively.

Proof: Let the eigenvalue decompostion of $\boldsymbol{S}$ be given by $\boldsymbol{S} = \boldsymbol{PDP}'$ and define $\boldsymbol{y} = \boldsymbol{P}'\boldsymbol{x}$. We find $\boldsymbol{y}'\boldsymbol{y} = \boldsymbol{x}'\boldsymbol{x}$, and

$$\frac{\boldsymbol{x}'\boldsymbol{Sx}}{\boldsymbol{x}'\boldsymbol{x}} = \frac{\boldsymbol{x}'\boldsymbol{PDP}'\boldsymbol{x}}{\boldsymbol{x}'\boldsymbol{x}} = \frac{\boldsymbol{y}'\boldsymbol{Dy}}{\boldsymbol{y}'\boldsymbol{y}} = \frac{\sum_{i=1}^n d_i y_i^2}{\sum_{i=1}^n y_i^2} = \sum_{i=1}^n w_i d_i, \tag{1}$$

where $w_i = y_i^2 / \sum_{i=1}^n y_i^2$ so that $0 \leq w_i \leq 1$ and $\sum_{i=1}^n w_i = 1$. Consequently, (1) describes convex combinations of the eigenvalues: the maximum equals the largest value, and the minimum equals the smallest eigenvalue. The extremes are reached by choosing for $\boldsymbol{x}$ the appropriate eigenvectors. q.e.d.

Notice that the squares of the singular values of the matrix $\boldsymbol{A}$, with SVD $\boldsymbol{A} = \boldsymbol{PDQ}'$ are eigenvalues of both $\boldsymbol{A}'\boldsymbol{A} = \boldsymbol{QD}^2\boldsymbol{Q}'$ and $\boldsymbol{AA}' = \boldsymbol{PD}^2\boldsymbol{P}'$, where $\boldsymbol{D}^2$ is diagonal with elements equal to the squares of the elements in $\boldsymbol{D}$.

4 Positive (semi)definite matrices

$\boldsymbol{S}$ is a *positive semidefinite* (psd) matrix if $\boldsymbol{S}$ is symmetric with nonnegative eigenvalues. In that case we write $\boldsymbol{S} \geq \mathbf{O}$. It does not mean that all elements of $\boldsymbol{S}$ are nonnegative. Sometimes the psd matrices are described as nonnegative definite.

Result 19

$$\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \geq \mathbf{O}$$

.

Exercise 7 *Prove Result 19*

$\boldsymbol{S}$ is *positive definite* pd, $\boldsymbol{S} > \mathbf{O}$, if $\boldsymbol{S}$ is symmetric with positive eigenvalues.

Result 20 *If $\boldsymbol{S}$ is nonsingular and $\boldsymbol{S} \geq \mathbf{O}$, then $\boldsymbol{S} > \mathbf{O}$.*

Exercise 8 *Prove Result 20.*

Result 21 $$\boldsymbol{S} \geq \mathbf{O} \Leftrightarrow \boldsymbol{x}'\boldsymbol{S}\boldsymbol{x} \geq 0, \ \forall \boldsymbol{x}.$$

As and alternative definition of postive semidefiniteness we can thus use $\boldsymbol{S} \geq \mathbf{O}$ if $\boldsymbol{x}'\boldsymbol{S}\boldsymbol{x} \geq 0$ for all conformable vectors $\boldsymbol{x}$. In the sequel we will use this definition. For an alternative definition of positive definiteness we can use the following result.

Result 22 $$\boldsymbol{S} > \mathbf{O} \Leftrightarrow \boldsymbol{x}'\boldsymbol{S}\boldsymbol{x} > 0, \ \forall \boldsymbol{x} \neq 0.$$

So, $\boldsymbol{S}$ will be said to be positive definite if $\boldsymbol{x}'\boldsymbol{S}\boldsymbol{x} > 0$ for all conformable vectors $\boldsymbol{x} \neq \mathbf{0}$.

Result 23 $$\boldsymbol{S} \geq \mathbf{O} \Leftrightarrow \boldsymbol{S} = \boldsymbol{A}\boldsymbol{A}' \ \ \textit{for some } \boldsymbol{A}.$$

Result 24 $$\boldsymbol{S} > \mathbf{O} \Leftrightarrow \boldsymbol{S} = \boldsymbol{C}\boldsymbol{C}' \textit{ for a nonsingular } \boldsymbol{C}.$$

Result 25 $$\boldsymbol{S} \geq \mathbf{O} \Rightarrow \boldsymbol{A}'\boldsymbol{S}\boldsymbol{A} \geq \mathbf{O}, \quad (\not\Leftarrow).$$

Exercise 9 *Prove Result 25.*

Result 26 $$\boldsymbol{S} > \mathbf{O} \Leftrightarrow \boldsymbol{C}'\boldsymbol{S}\boldsymbol{C} > \mathbf{O} \textit{ for a nonsingular } \boldsymbol{C}.$$

Notice that in Results 23–26 $\boldsymbol{A}$ and $\boldsymbol{C}$ can be replaced by (conformable) $\boldsymbol{A}'$ and $\boldsymbol{C}'$.
In particular it follows from Result 21, or 25, that the diagonal elements of a psd matrix are nonnegative: $\boldsymbol{e}_i'\boldsymbol{S}\boldsymbol{e}_i \geq 0$. For a pd matrix the diagonal elements are positive as follow from the next result.

Result 27 $$\boldsymbol{S} > \mathbf{O} \Rightarrow \boldsymbol{A}'\boldsymbol{S}\boldsymbol{A} > \mathbf{O} \textit{ if } \boldsymbol{A} \textit{ has full column rank.}$$

Exercise 10 *Prove Result 27.*

By using $\boldsymbol{C} = \boldsymbol{S}^{-1}$ in Result 26 we find the following result.

Result 28 $$\boldsymbol{S} > \mathbf{O} \Leftrightarrow \boldsymbol{S}^{-1} > \mathbf{O}.$$
A more general result is given without proof.

Result 29 $$\boldsymbol{S} \geq \boldsymbol{C} > \mathbf{O} \Leftrightarrow \boldsymbol{C}^{-1} \geq \boldsymbol{S}^{-1} > \mathbf{O}.$$

Let $\boldsymbol{D}$ be diagonal and psd. For positive values of the scalar k the matix $\boldsymbol{D}^k$ is defined as the diagonal matrix with d_i^k as diagonal elements. If $\boldsymbol{D} > \mathbf{O}$, then $\boldsymbol{D}^k$ can thus be defined for negative values of k as well. Let $\boldsymbol{S} \geq \mathbf{O}$ with eigenvalue decomposition $\boldsymbol{S} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{P}'$, then define for $k > 0$
$$\boldsymbol{S}^k = \boldsymbol{P}\boldsymbol{D}^k\boldsymbol{P}'.$$

For $\boldsymbol{S} > \mathbf{O}$ this definition also applies for negative values of k.

Result 30 $$\boldsymbol{S} \geq \mathbf{O} \Rightarrow \boldsymbol{S} = \boldsymbol{S}^{1/2}\boldsymbol{S}^{1/2}.$$
(Compare with Result 23)

Result 31 $$\boldsymbol{S} > \mathbf{O} \Rightarrow \boldsymbol{S}^{-1} = \boldsymbol{S}^{-1/2}\boldsymbol{S}^{-1/2}.$$

5 Projection matrices

The matrix $\boldsymbol{B}$ is *idempotent* if $\boldsymbol{B} = \boldsymbol{B}\boldsymbol{B}$. So, an idempotent matrix is square, but not necessarily symmetric. That is, if $\boldsymbol{B}$ is idempotent, then so is $\boldsymbol{C}^{-1}\boldsymbol{B}\boldsymbol{C}$.

$\boldsymbol{Q}$ is a *projection matrix* if it is idempotent and symmetric, which will be indicated by $\boldsymbol{Q} \in \mathcal{P}$.

Result 32 $$\boldsymbol{Q} \in \mathcal{P} \Rightarrow \boldsymbol{Q} \geq \mathbf{O}.$$

Exercise 11 *Prove Result 32.*

Result 33 *If $\boldsymbol{Q} \in \mathcal{P}$, then the eigenvalues of $\boldsymbol{Q}$ are equal to either zero or one.*

Exercise 12 *Prove Result 33.*

Result 34 $$\boldsymbol{Q} \in \mathcal{P} \Leftrightarrow \boldsymbol{Q} = \boldsymbol{P}\boldsymbol{P}' \text{ for an orthonormal } \boldsymbol{P}.$$

Let $\boldsymbol{A}$ be an $n \times m$ matrix of full column rank. Then, by Result 11, $\boldsymbol{A}'\boldsymbol{A}$ is nonsingular, and by Result 23, $\boldsymbol{A}'\boldsymbol{A} \geq \mathbf{O}$. So, by Result 20, $\boldsymbol{A}'\boldsymbol{A} > \mathbf{O}$ and $(\boldsymbol{A}'\boldsymbol{A})^{-1/2}$ exists.

Result 35 *If $\boldsymbol{A}$ has full column rank, then $\boldsymbol{A}(\boldsymbol{A}'\boldsymbol{A})^{-1/2}$ is orthonormal.*

Result 36 *If* $\boldsymbol{A}$ *has full column rank and* $\boldsymbol{P}_A = \boldsymbol{A}(\boldsymbol{A}'\boldsymbol{A})^{-1}\boldsymbol{A}'$*, then* $\boldsymbol{P}_A \in \mathcal{P}$*.*

The last result follows from Results 34 and 35. Alternatively, it can be verified that $\boldsymbol{P}_A$ is symmetric and idempotent and, consequently, a projection matrix.

Notice that $\boldsymbol{I} - \boldsymbol{P}_A$ is a projection matrix as well. That is, $(\boldsymbol{I} - \boldsymbol{P}_A)(\boldsymbol{I} - \boldsymbol{P}_A) = \boldsymbol{I} - \boldsymbol{P}_A$ and $\boldsymbol{I} - \boldsymbol{P}_A$ is symmetric.

Result 37 $\qquad \boldsymbol{I} - \boldsymbol{P}_A \geq \mathbf{O}, \ \textit{or}\ \boldsymbol{P}_A \leq \boldsymbol{I}.$

(It follows from Result 32.)

So, $0 \leq \boldsymbol{P}_A \leq \boldsymbol{I}$. Thus we have $\boldsymbol{P}_A\boldsymbol{A} = \boldsymbol{A}$, $(\boldsymbol{I} - \boldsymbol{P}_A)\boldsymbol{A} = \mathbf{O}$, $(\boldsymbol{I} - \boldsymbol{P}_A)\boldsymbol{P}_A = \mathbf{O}$. Consequently, a vector $\boldsymbol{z}$ can be decomposed in a vector formed by a linear combination of the columns of $\boldsymbol{A}$, i.e. a vector in the range space of $\boldsymbol{A}$, to wit $\boldsymbol{P}_A\boldsymbol{z}$, and a vector orthogonal to the first component:

$$\boldsymbol{z} = \boldsymbol{P}_A\boldsymbol{z} + (\boldsymbol{I} - \boldsymbol{P}_A)\boldsymbol{z}.$$

We now have analogous to the Pythagorean theorem the following result.

Result 38 $\qquad \|\boldsymbol{z}\|^2 = \|\boldsymbol{P}_A\boldsymbol{z}\|^2 + \|(\boldsymbol{I} - \boldsymbol{P}_A)\boldsymbol{z}\|^2.$

A frequently used projection matrix is given by

$$\boldsymbol{P}_{\imath_n} = \boldsymbol{\imath}_n(\boldsymbol{\imath}'\boldsymbol{\imath})^{-1}\boldsymbol{\imath}' = \frac{\boldsymbol{\imath}\boldsymbol{\imath}'}{\boldsymbol{\imath}'\boldsymbol{\imath}} = \frac{\boldsymbol{\imath}\boldsymbol{\imath}'}{n} = n^{-1}\begin{bmatrix} 1 & 1 & \cdots & 1 \\ 1 & 1 & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 1 \end{bmatrix}.$$

So, $\boldsymbol{P}_{\imath_n}$ is a matrix of rank one with all elements equal to n^{-1}. Let

$$\bar{z} = n^{-1}\sum_{i=1}^{n} z_i = \frac{\boldsymbol{\imath}'\boldsymbol{z}}{\boldsymbol{\imath}'\boldsymbol{\imath}},$$

then

$$\begin{aligned} \boldsymbol{P}_{\imath_n}\boldsymbol{z} &= \tfrac{\boldsymbol{\imath}\boldsymbol{\imath}'}{\boldsymbol{\imath}'\boldsymbol{\imath}}\boldsymbol{z} = \boldsymbol{\imath}_n\bar{z}, \\ (\boldsymbol{I} - \boldsymbol{P}_{\imath_n})\boldsymbol{z} &= \boldsymbol{z} - \boldsymbol{\imath}_n\bar{z} = \begin{pmatrix} z_1 - \bar{z} \\ \vdots \\ z_n - \bar{z} \end{pmatrix}. \end{aligned}$$

The last vector contains the elements of $\boldsymbol{z}$ in deviation from the mean.

Exercise 13 *Prove if both $\boldsymbol{A}$ and $\boldsymbol{AB}$ have full column rank, then*

$$\boldsymbol{P_A} - \boldsymbol{P}_{\{\boldsymbol{AB}\}} \geq 0.$$

6 Scalar functions of square matrices

The *trace* of an $n \times n$ matrix $\boldsymbol{C}$ is given by the sum of its diagonal elements: $\mathrm{tr}(\boldsymbol{C}) = \sum_{i=1}^{n} c_{ii}$.

Result 39 $$\mathrm{tr}(\boldsymbol{AB}) = \mathrm{tr}(\boldsymbol{BA}).$$

Result 40 $$\boldsymbol{x}'\boldsymbol{Ay} = \mathrm{tr}(\boldsymbol{Ayx}').$$

Exercise 14 *Prove Result 40.*

Result 41 $$\boldsymbol{Q} \in \mathcal{P} \Rightarrow \mathrm{tr}(\boldsymbol{Q}) = \mathrm{rank}(\boldsymbol{Q}).$$

Proof: By Result 34 we find $\boldsymbol{Q} = \boldsymbol{PP}'$, for an orthonormal $n \times r$ matrix $\boldsymbol{P}$. It follows from Result 12 that $\mathrm{rank}(\boldsymbol{Q}) = \mathrm{rank}(\boldsymbol{P}) = r$, and by Result 39 we find $\mathrm{tr}(\boldsymbol{Q}) = \mathrm{tr}(\boldsymbol{PP}') = \mathrm{tr}(\boldsymbol{P}'\boldsymbol{P}) = \mathrm{tr}(\boldsymbol{I}_r) = r$. q.e.d.

Result 42 *If the $n \times m$ matrix $\boldsymbol{A}$ has full column rank m and $\boldsymbol{P}_A = \boldsymbol{A}(\boldsymbol{A}'\boldsymbol{A})^{-1}\boldsymbol{A}'$, then*

$$\begin{aligned} \mathrm{tr}(\boldsymbol{P}_A) &= m, \\ \mathrm{tr}(\boldsymbol{I}_n - \boldsymbol{P}_A) &= n - m. \end{aligned}$$

Result 43 *If $\boldsymbol{S}$ is symmetric, then $\mathrm{tr}(\boldsymbol{S})$ equals the sum of eigenvalues of $\boldsymbol{S}$.*

Exercise 15 *Prove Result 43.*

If $\boldsymbol{S}$ is symmetric, then the *determinant* of $\boldsymbol{S}$, $|\boldsymbol{S}|$, is given by the product of its eigenvalues.

Result 44 $$|\boldsymbol{S}^{-1}| = (|\boldsymbol{S}|)^{-1}.$$

Result 45 *If α is a scalar and $\boldsymbol{S}$ is $n \times n$, then $|\alpha\boldsymbol{S}| = |\alpha\boldsymbol{I}_n\boldsymbol{S}| = |\alpha\boldsymbol{I}_n||\boldsymbol{S}| = \alpha^n|\boldsymbol{S}|$.*

7 Partitioned matrices

Let $\boldsymbol{A}$ and $\boldsymbol{B}$ be given by

$$\boldsymbol{A} = \begin{bmatrix} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22} \end{bmatrix}, \quad \boldsymbol{B} = \begin{bmatrix} \boldsymbol{B}_{11} & \boldsymbol{B}_{12} \\ \boldsymbol{B}_{21} & \boldsymbol{B}_{22} \end{bmatrix},$$

where the submatrices $\boldsymbol{A}_{ij}$ are of the same order as $\boldsymbol{B}_{ij}$, $i, j = 1, 2$.

Result 46
$$\boldsymbol{A} + \boldsymbol{B} = \begin{bmatrix} \boldsymbol{A}_{11} + \boldsymbol{B}_{11} & \boldsymbol{A}_{12} + \boldsymbol{B}_{12} \\ \boldsymbol{A}_{21} + \boldsymbol{B}_{21} & \boldsymbol{A}_{22} + \boldsymbol{B}_{22} \end{bmatrix}.$$

Result 47
$$\boldsymbol{A}'\boldsymbol{B} = \begin{bmatrix} \boldsymbol{A}'_{11}\boldsymbol{B}_{11} + \boldsymbol{A}'_{21}\boldsymbol{B}_{21} & \boldsymbol{A}'_{11}\boldsymbol{B}_{12} + \boldsymbol{A}'_{21}\boldsymbol{B}_{22} \\ \boldsymbol{A}'_{12}\boldsymbol{B}_{11} + \boldsymbol{A}'_{22}\boldsymbol{B}_{21} & \boldsymbol{A}'_{12}\boldsymbol{B}_{12} + \boldsymbol{A}'_{22}\boldsymbol{B}_{22} \end{bmatrix}.$$

Let the matrix $\boldsymbol{S}$ be given by

$$\boldsymbol{S} = \begin{bmatrix} \boldsymbol{S}_{11} & \boldsymbol{S}_{12} \\ \boldsymbol{S}_{21} & \boldsymbol{S}_{22} \end{bmatrix},$$

where $\boldsymbol{S}_{11}$ and $\boldsymbol{S}_{22}$ are square $n \times n$ and $m \times m$, respectively.

Result 48
$$\boldsymbol{S} > \mathbf{O} \Rightarrow \boldsymbol{S}_{11} > \mathbf{O} \textit{ and } \boldsymbol{S}_{22} > \mathbf{O}.$$

Exercise 16 *Prove Result 48.*

Result 49 *If* $\boldsymbol{S}_{12} = \mathbf{O}$*, i.e.* $\boldsymbol{S}$ *is* block-triangular*, and* $\boldsymbol{S}_{11}$ *and* $\boldsymbol{S}_{22}$ *are nonsingular, then*

$$\boldsymbol{S}^{-1} = \begin{bmatrix} \boldsymbol{S}_{11}^{-1} & \mathbf{O} \\ -\boldsymbol{S}_{22}^{-1}\boldsymbol{S}_{21}\boldsymbol{S}_{11}^{-1} & \boldsymbol{S}_{22}^{-1} \end{bmatrix}.$$

So, if both $\boldsymbol{S}_{21}$ and $\boldsymbol{S}_{12}$ consist of zeros, i.e. $\boldsymbol{S}$ is *block-diagonal*, then $\boldsymbol{S}^{-1}$ is block-diagonal as well.

Result 50 *In general, for a partitioned matrix* S*, we find*

$$\begin{aligned} \boldsymbol{S} &= \begin{bmatrix} \boldsymbol{I}_n & \boldsymbol{S}_{12}\boldsymbol{S}_{22}^{-1} \\ \mathbf{O} & \boldsymbol{I}_m \end{bmatrix} \begin{bmatrix} \boldsymbol{S}_{11} - \boldsymbol{S}_{12}\boldsymbol{S}_{22}^{-1}\boldsymbol{S}_{21} & \mathbf{O} \\ \mathbf{O} & \boldsymbol{S}_{22} \end{bmatrix} \begin{bmatrix} \boldsymbol{I}_n & \mathbf{O} \\ \boldsymbol{S}_{22}^{-1}\boldsymbol{S}_{21} & \boldsymbol{I}_m \end{bmatrix} \\ &= \begin{bmatrix} \boldsymbol{S}_{11} - \boldsymbol{S}_{12}\boldsymbol{S}_{22}^{-1}\boldsymbol{S}_{21} & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{bmatrix} + \begin{bmatrix} \boldsymbol{S}_{12}\boldsymbol{S}_{22}^{-1} \\ \boldsymbol{I}_m \end{bmatrix} \boldsymbol{S}_{22} \begin{bmatrix} \boldsymbol{S}_{22}^{-1}\boldsymbol{S}_{21} & \boldsymbol{I}_m \end{bmatrix} \end{aligned}$$

Exercise 17 *Let* $\boldsymbol{S} > 0$*, verify* $\boldsymbol{S}_{11} - \boldsymbol{S}_{12}\boldsymbol{S}_{22}^{-1}\boldsymbol{S}_{21} > 0$.

Exercise 18 *Let $\boldsymbol{S} > 0$. Find the $m \times n$ matrix $\boldsymbol{A}$ that satisfies*

$$\boldsymbol{A} = \arg\min_{\boldsymbol{A}} \left\{ \begin{bmatrix} \boldsymbol{I}_n & -\boldsymbol{A}' \end{bmatrix} \boldsymbol{S} \begin{bmatrix} \boldsymbol{I}_n \\ -\boldsymbol{A} \end{bmatrix} \right\},$$

where the matrix has a minimum in the p.s.d. sense.

Result 51 *In general, for a nonsingular partitioned matrix $\boldsymbol{S}$, we find*

$$\begin{aligned} \boldsymbol{S}^{-1} &= \begin{bmatrix} \boldsymbol{I}_n & \mathbf{O} \\ -\boldsymbol{S}_{22}^{-1}\boldsymbol{S}_{21} & \boldsymbol{I}_m \end{bmatrix} \begin{bmatrix} (\boldsymbol{S}_{11} - \boldsymbol{S}_{12}\boldsymbol{S}_{22}^{-1}\boldsymbol{S}_{21})^{-1} & \mathbf{O} \\ \mathbf{O} & \boldsymbol{S}_{22}^{-1} \end{bmatrix} \begin{bmatrix} \boldsymbol{I}_n & -\boldsymbol{S}_{12}\boldsymbol{S}_{22}^{-1} \\ \mathbf{O} & \boldsymbol{I}_m \end{bmatrix} \\ &= \begin{bmatrix} \boldsymbol{I}_n \\ -\boldsymbol{S}_{22}^{-1}\boldsymbol{S}_{21} \end{bmatrix} (\boldsymbol{S}_{11} - \boldsymbol{S}_{12}\boldsymbol{S}_{22}^{-1}\boldsymbol{S}_{21})^{-1} \begin{bmatrix} \boldsymbol{I}_n & -\boldsymbol{S}_{12}\boldsymbol{S}_{22}^{-1} \end{bmatrix} + \begin{bmatrix} \mathbf{O} & \mathbf{O} \\ \mathbf{O} & \boldsymbol{S}_{22}^{-1} \end{bmatrix}. \end{aligned}$$

So,

$$(\boldsymbol{I}_n, \mathbf{O})\boldsymbol{S}^{-1} \begin{pmatrix} \boldsymbol{I}_n \\ \mathbf{O} \end{pmatrix} = \left(\boldsymbol{S}_{11} - \boldsymbol{S}_{12}\boldsymbol{S}_{22}^{-1}\boldsymbol{S}_{21}\right)^{-1}. \tag{2}$$

8 Matrix differentiation

Let $\boldsymbol{x}$ be an m-vector and the function $\boldsymbol{f}(\boldsymbol{x})$ an n-vector. The partial derivatives can be arranged in a matrix as follows.

$$\frac{\partial \boldsymbol{f}(\boldsymbol{x})}{\partial \boldsymbol{x}'} = \frac{\partial \begin{pmatrix} f_1(\boldsymbol{x}) \\ \vdots \\ f_n(\boldsymbol{x}) \end{pmatrix}}{\partial (x_1, \ldots, x_m)} = \begin{bmatrix} \frac{\partial f_1(\boldsymbol{x})}{\partial x_1} & \cdots & \frac{\partial f_1(\boldsymbol{x})}{\partial x_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n(\boldsymbol{x})}{\partial x_1} & \cdots & \frac{\partial f_n(\boldsymbol{x})}{\partial x_m} \end{bmatrix}.$$

We can now use the chain rule. Let $\boldsymbol{x} = \boldsymbol{x}(\boldsymbol{\theta})$, then

$$\frac{\partial \boldsymbol{f}(\boldsymbol{x}(\boldsymbol{\theta}))}{\partial \theta'} = \frac{\partial \boldsymbol{f}(\boldsymbol{x})}{\partial \boldsymbol{x}'} \frac{\partial \boldsymbol{x}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}'}.$$

Result 52

$$\frac{\partial \boldsymbol{A}\boldsymbol{x}}{\partial \boldsymbol{x}'} = \boldsymbol{A}.$$

Result 53

$$\frac{\partial \boldsymbol{x}'\boldsymbol{A}\boldsymbol{x}}{\partial \boldsymbol{x}'} = \boldsymbol{x}'(\boldsymbol{A} + \boldsymbol{A}').$$

Sometimes it convenient to differentiate a scalar function of a matrix with respect to the elements of that matrix. Let the matrix of derivatives have elements

$$\left\{\frac{\partial f(\boldsymbol{A})}{\partial \boldsymbol{A}}\right\}_{ij} = \frac{\partial f(\boldsymbol{A})}{\partial A_{ij}},$$

so it has the same order as $\boldsymbol{A}$.

Result 54 *If $\boldsymbol{A}$ is $m \times m$, then*

$$\frac{\partial \operatorname{tr} \boldsymbol{A}}{\partial \boldsymbol{A}} = \boldsymbol{I}_m.$$

Result 55

$$\frac{\partial |\boldsymbol{A}|}{\partial \boldsymbol{A}} = |\boldsymbol{A}|\boldsymbol{A}'^{-1}.$$

Exercise 19 *Let $\boldsymbol{S} > 0$ be a $(1+m) \times (1+m)$ matrix. Find by differentiation the m vector $\boldsymbol{x}$ that satisfies*

$$\boldsymbol{x} = \arg\min_{\boldsymbol{x}} \left\{ \begin{pmatrix} 1 & -\boldsymbol{x}' \end{pmatrix} \boldsymbol{S} \begin{pmatrix} 1 \\ -\boldsymbol{x} \end{pmatrix} \right\}.$$

Further reading: Abadir, K.M., and J.R. Magnus, 2005: *Matrix Algebra*, Cambridge University Press.